

D02-2: PWM controller

Pulse Width Modulation

Generate a square wave with a configurable duty cycle.

The average DC voltage drives an electric motor.

Interface signals

CLK : Core clock

RST : Active-high reset

PERIOD : Number of pulse period

DUTY : Number of ON cycles

SET : Set DUTY and PERIOD, enable PWM

STOP : Stop the PWM output

PWM : Generated PWM pulse

RDY : Active-low when the PWM core is activated

Detailed operations

Reset the core (RDY becomes HIGH).

Set PERIOD and DUTY.

Assert 1-cycle SET (active-high) to enable the PWM core.

Assert 1-cycle STOP (active-high) to stop the PWM core.

A state machine keeps the core in IDLE until it is set up, loads the settings for one cycle, and runs until it is stopped.